

Project Name:

TriLit AI - An All-in-One Research Literature Assistant

Student Name:

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1. Technical Description

TriLit AI is an NLP (Natural Language Processing) powered desktop tool designed to help researchers, PhD students, and academics manage the overwhelming task of literature review. The tool combines three core functions into a single, easy-to-use interface.

How It Works (The Three Modules):

Smart Paper Discovery (ScholarSort): The user types in a research question or topic (e.g., "How does BERT improve medical text classification?"). The system uses a technique called "semantic search" to scan a large database of academic papers (like arXiv or PubMed). Instead of just matching keywords, it understands the meaning of the query and returns a list of the most relevant papers, automatically organizing them by sub-field.

Instant Paper Digest (PaperSummarizer): Once the user selects a paper, the system generates a short, plain-English summary. Unlike a standard abstract, this summary highlights the paper's unique contribution - specifically what new method or finding the paper introduces. This allows the user to decide in under one minute whether the paper is worth a full read.

Automated Literature Review (LitRev-AI): For systematic reviews or thesis chapters, the user can upload a collection of papers. The AI screens each paper based on a simple "Include/Exclude" logic defined by the user's research criteria. It then exports a clean list of selected papers and automatically generates a PRISMA Flow Diagram, which is a standard requirement for publishing systematic reviews.

Technical Implementation (No Paid APIs Required):

The entire system runs locally on a standard computer without needing expensive cloud API keys or internet credits. It uses open-source machine learning libraries (Hugging Face Transformers, PyTorch, FAISS). The models used are lightweight and efficient versions of state-of-the-art architectures (e.g., DistilBERT, BART), ensuring that the tool can function on a standard university laptop.

2. Business Description & Market Value

TriLit AI is positioned as a Software-as-a-Service (SaaS) product aimed at the global academic research community.

The Problem We Solve:

Currently, researchers waste 6 to 10 hours per week on manual, repetitive tasks: searching databases, reading irrelevant abstracts, and manually formatting citation lists for literature reviews. Generic AI tools (like ChatGPT) are not reliable for academic work because they often "hallucinate" (invent) fake citations and cannot follow strict academic screening protocols like PRISMA.

Target Market:

Primary Users: Individual graduate students, PhD candidates, and postdoctoral researchers.

Secondary Users: University libraries and research labs seeking institutional licenses for their members.

Market Opportunity:

The global market for AI in academic research is expected to reach \$5-7 billion by 2028, growing at over 20% annually. This growth is driven by the increasing volume of publications (over 3 million new papers published each year) and pressure on researchers to publish faster.

Value Proposition:

TriLit AI saves time and increases accuracy. By reducing literature review time by an estimated 50-70%, the tool allows researchers to focus on actual experimentation and

writing. For universities, this translates to higher research output and improved grant application success rates.

- Freemium Tier: Free access to basic search and limited summaries (attracts users).
 - Pro Tier (\$9.99/month): Unlimited papers, advanced filtering, and export options.
 - Institutional License: Custom pricing for university-wide access to the PRISMA generator and team collaboration features.
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